

Comprehensive Game Economy Rebalancing and Systems Audit: Mystic Motors

1. Macroeconomic Strategy and Trophy-Milestone Framework

The digital economy of "Mystic Motors" relies on a highly retentive three-queue timer system (Car Upgrades, Spell Research, and Crate Opening).¹ To ensure long-term retention without alienating the free-to-play (F2P) base, this rebalancing completely abandons arbitrary scaling in favor of a **Trophy-Milestone-Based Balancing Framework**.

Under this model, a player's expected car level, spell level, and tier are strictly correlated with their competitive Trophy count. All upgrade costs, shard requirements, and XP gates are reverse-engineered from the player's expected earnings per race at their specific matchmaking rank.¹

The economy maintains a strict anchor of \$1.00 USD to 100 Gems.¹ Furthermore, Mastery XP (earned purely through racing) remains the uncompromising gate for progression, meaning the highest spenders cannot simply buy their way into late-game tiers without participating in the competitive ecosystem.¹

The Cost-to-Earnings Ratio Framework:

All Soft Currency (Coin) costs are generated dynamically based on the formula:

$$item_cost = avg_earning_per_race_at_expected_rank \times target_races_to_afford$$

- **Low levels (0-3):** Require 1-5 races' worth of coins.
- **Mid levels (4-6):** Require 5-15 races' worth of coins.
- **High levels (7-9):** Require 15-40 races' worth of coins.

2. Systemic Remediation of Audit Flags

2.1 Complete Removal of Spell Tokens

The legacy "Spell Token" system has been entirely excised from the economy. The V2 system relies exclusively on **Spell Shards** (earned at a rate of 4 to 30 per race based on rank, plus crate drops).¹ Spells are now unlocked and leveled using only Spell Shards and research timers, with a maximum of 5 levels per spell.

2.2 League Rewards and SKU Corrections

The legacy "V1 Key" SKUs have been purged from all loot tables. No custom tokens (e.g., Chrono/Catalyst) have been invented. Instead, League Rewards have been modernized to inject surgical bursts of existing economy assets:

- **Direct Currency:** Coins, Gems, and Spell Shards.
- **Time Speed-ups:** Ranges from 5m, 15m, 1h, 4h, 12h, 24h, up to 7d.
- **Crates:** Utilizing native SKUs exclusively (sku_zz3twgp0wx for Common, sku_n9hsc0wxxk for Rare, sku_kgkjadrd79 for Exotic, etc.).¹

2.3 Resolving the Mastery Rank 30 Threshold Bug

The legacy configuration files contained a progression bug breaking the transition between Mastery Rank 29 and Rank 30. This has been resolved by implementing a strictly monotonic formula for all 50 ranks. The tier gates have been correctly mapped to Ranks 5, 10, 20, and 30 to support a rapid early-game hook followed by a deep mid-game curve.

2.4 Timer Normalization and Cap

The longest timer in the game is strictly capped at **48 hours**. Previous iterations featuring 21-day timers were fundamentally incompatible with mobile arcade racing retention. The new pacing triggers frequent session returns (15m to 4h in the early game) and scales to a maximum of 2 days for elite endgame upgrades, matching established market standards like Clash Royale.

2.5 Unification of Premium Skip Costs

Timer skip costs now follow a unified, diminishing-returns formula. At the hard anchor of \$1 = 100 Gems, the formula heavily penalizes 5-minute skips while providing scalable value for skipping 24h or 48h timers, ensuring Whale players feel they are receiving premium value in the late game.

3. Core Progression Formulas

All progression variables have been run through a global $\text{roundNice}(\text{value})$ algorithm to ensure clean, player-facing numbers (snapping to the nearest 50, 100, 500, or 1000 depending on magnitude).

- **Car XP Requirement:**
$$\text{carXpForLevel}(\text{tier}, \text{level}) = \text{roundNice}(\text{BaseXP}[\text{tier}] \times \text{level}^{1.5})$$
- **Car Coin Cost:**
$$\text{carCoinCost}(\text{tier}, \text{level}) = \text{target_races}(\text{level}) \times \text{ExpectedEarning}[\text{tier}]$$
- **Mastery Threshold:**

$$masteryThreshold(rank) = roundNice(3500 \times rank^{1.35})$$

- **Gem Skip Cost:** $gemSkipCost(hours) = roundNice(25 \times hours^{0.85})$

4. Rebalanced Progression Tables

4.1 Car Coin Costs (By Expected Earnings)

Costs perfectly align with the Trophy-Milestone framework. A Tier 3 upgrade at level 7 requires 20 races at Platinum/Champion earning levels (~12,000 coins/race).

Level Transition	Target Races	Tier 1 (Base 2.5k)	Tier 2 (Base 4k)	Tier 3 (Base 8k)	Tier 4 (Base 15k)	Tier 5 (Base 25k)
Level 0 to 1	1	2,500	4,000	8,000	15,000	25,000
Level 1 to 2	2	5,000	8,000	16,000	30,000	50,000
Level 2 to 3	4	10,000	16,000	32,000	60,000	100,000
Level 3 to 4	6	15,000	24,000	48,000	90,000	150,000
Level 4 to 5	10	25,000	40,000	80,000	150,000	250,000
Level 5 to 6	15	40,000	60,000	120,000	225,000	375,000
Level 6 to 7	20	50,000	80,000	160,000	300,000	500,000

Level 7 to 8	25	65,000	100,000	200,000	375,000	625,000
Level 8 to 9	30	75,000	120,000	240,000	450,000	750,000
Level 9 to 10	40	100,000	160,000	320,000	600,000	1,000,000

4.2 Car Experience Requirements

Experience costs scale to prevent late-game grinding walls while enforcing gameplay engagement.¹

Level Transition	Tier 1	Tier 2	Tier 3	Tier 4	Tier 5
Level 0 to 1	100	250	500	1,000	2,000
Level 1 to 2	200	400	800	1,500	3,000
Level 2 to 3	300	600	1,200	2,000	4,000
Level 3 to 4	400	800	1,600	3,000	6,000
Level 4 to 5	500	1,000	2,000	4,000	8,000
Level 5 to 6	600	1,500	2,500	5,000	10,000

Level 6 to 7	800	2,000	3,000	6,000	12,000
Level 7 to 8	1,000	2,500	4,000	8,000	16,000
Level 8 to 9	1,200	3,000	5,000	10,000	20,000
Level 9 to 10	1,500	4,000	6,000	12,000	25,000

4.3 Car Upgrade Timers

Timers are clamped to ensure returning sessions fit into healthy daily routines (morning/evening) without stretching into absurdity.

Level Transition	Tier 1	Tier 2	Tier 3	Tier 4	Tier 5
Level 0 to 1	1 Min	15 Mins	1 Hour	2 Hours	4 Hours
Level 1 to 2	2 Mins	30 Mins	2 Hours	4 Hours	8 Hours
Level 2 to 3	5 Mins	45 Mins	3 Hours	6 Hours	12 Hours
Level 3 to 4	10 Mins	1 Hour	4 Hours	8 Hours	16 Hours
Level 4 to 5	15 Mins	1.5 Hours	5 Hours	10 Hours	20 Hours

Level 5 to 6	20 Mins	2 Hours	6 Hours	12 Hours	24 Hours
Level 6 to 7	25 Mins	2.5 Hours	8 Hours	16 Hours	32 Hours
Level 7 to 8	30 Mins	3 Hours	10 Hours	20 Hours	40 Hours
Level 8 to 9	30 Mins	4 Hours	12 Hours	24 Hours	48 Hours
Level 9 to 10	30 Mins	4 Hours	12 Hours	24 Hours	48 Hours

4.4 Spell Research Dependencies (5 Levels)

Spells run exclusively on Shards and cap at Level 5.

Level Transition	Shard Cost	Target Trophy Range / Shard Income	Research Timer
Level 0 to 1	100	0-1,500 (Earns ~8/race)	1 Hour
Level 1 to 2	300	1,500-3,000 (Earns ~13/race)	4 Hours
Level 2 to 3	1,000	3,000-5,000 (Earns ~18/race)	12 Hours

Level 3 to 4	2,500	5,000-7,000 (Earns ~25/race)	24 Hours
Level 4 to 5	5,000	7,000+ (Earns ~30/race)	48 Hours

4.5 Mastery Rank Thresholds and Tier Unlock Costs

Mastery XP accumulates at approximately 157 per race.¹ The Tier 2 gate opens cleanly at Rank 5, perfectly pacing the early game hook. Tier unlock costs are priced at 20-40 races worth of expected earnings at that rank.

Progression Milestone	Target Rank	Cumulative XP Req	Trophy / Earnings Target	Tier Unlock Cost (Coins)
Account Creation	Rank 1	0	Unranked	N/A (Tier 1 Default)
Tier 2 Vehicle Unlock	Rank 5	30,500	Silver (~3,500/race)	100,000
Tier 3 Vehicle Unlock	Rank 10	78,000	Platinum (~6,000/race)	180,000
Tier 4 Vehicle Unlock	Rank 20	200,000	Champion (~10,000/race)	300,000
Tier 5 Vehicle Unlock	Rank 30	343,000	Hypersonic (~25,000/race)	750,000

5. Car Stat Diversification and Archetype Budgets

The legacy placeholder of identical handling stats across all vehicles has been corrected. Vehicles now allocate their performance across all 5 stats (topSpeed, acceleration, handling, boostRegen, boostPower).

To maintain balance, the **Total Stat Budget** is equalized within each tier, but min/max ranges are aggressively enforced to guarantee unique archetype "feels."

- **Tier 1 Total Budget:** 300
- **Tier 2 Total Budget:** 400
- **Tier 3 Total Budget:** 500
- **Tier 4 Total Budget:** 600
- **Tier 5 Total Budget:** 700

Archetype Min/Max Ranges (Example based on Tier 1 Budget of 300):

To calculate higher tiers, add +20 to all stats per Tier increase.

Archetype	Top Speed	Acceleration	Handling	Boost Regen	Boost Power
Tank	75 - 85	35 - 45	30 - 40	50 - 60	85 - 95
Speedster	80 - 90	80 - 90	40 - 50	40 - 50	35 - 45
Specialist	45 - 55	55 - 65	80 - 90	65 - 75	30 - 40

6. Daily Income and Expense Modeling by Profile

Based on the 62.8% crate drop chance and the Elo-style trophy progression ¹, player journeys synchronize their resource influx with their mechanical ceilings.

6.1 The Casual Player Model (Bronze/Silver Elo)

- **Engagement:** 5 races/day, 1 session (15 mins).
- **Daily Income:** ~15,000 Coins, ~785 Mastery XP, ~40 Shards.
- **Economy State:** Casual players clear Tier 1 seamlessly. Because their timers are short

(1-30 mins), they complete upgrades within their session. They hit Rank 5 and unlock Tier 2 in roughly 5 weeks. Their minimal playstyle means they won't feel resource starvation until deeply into Tier 2.

6.2 The Regular F2P Player Model (Platinum/Diamond Elo)

- **Engagement:** 10 races/day, 3 sessions (30 mins).
- **Daily Income:** ~80,000 Coins, ~1,570 Mastery XP, ~130 Shards.
- **Economy State:** Achieves Rank 5 in ~20 days. Progression perfectly tracks with Trophies. By the time they reach Tier 3, their 80,000 daily coin income aligns beautifully with the 32,000 to 80,000 coin costs of mid-level upgrades. They return 3 times daily to claim 2-to-6 hour timers.

6.3 The Hardcore Player Model (Master/Champion Elo)

- **Engagement:** 20 races/day, 5 sessions (60 mins).
- **Daily Income:** ~300,000 Coins, ~3,140 Mastery XP, ~400 Shards.
- **Economy State:** Hits the Rank 10 (Tier 3) gate in roughly 25 days. Their volume ensures they never suffer a Soft Currency deficit, allowing them to fund multiple vehicle upgrades simultaneously. Their true bottleneck is the **XP Gate**—they have the money, but must actively grind the races to unlock the next level nodes.

6.4 The Whale Player Model (Hypersonic Elo)

- **Engagement:** 20 races/day + \$100/mo spend.
- **Daily Income:** Unlimited via Shop.
- **Economy State:** Uses Gems (100 Gems = \$1) to skip 12h-48h endgame timers. Skipping a 24-hour Tier 4 timer costs roughly 360 Gems (\$3.60). However, because they are hard-capped by the unskippable Mastery formula, the Whale is forced to play 218 days (~7 months) of active, high-level matchmaking to reach Rank 30 and unlock Tier 5, ensuring they provide content and competition for the F2P base.

7. Monetization Pressure Analysis

By correlating costs to expected earnings at specific Trophy brackets, the game naturally applies monetization pressure without ever making the player feel artificially "robbed" of progress.

- **Early Game (0-1500 Trophies): The Impatience Hook.**
Players have abundant coins but must wait 30-45 minutes. A \$4.99 "Starter Bundle" offering 500 Gems and a dozen 15-minute Speed-Up Boosters converts them by providing instant gratification.
- **Mid Game (3000-5000 Trophies): The Horizon Sale.**
Players hit 12-hour timers. Here, the game offers high-value "Season Passes." The player sees a 12-hour timer and realizes it perfectly bridges their evening session to their

morning session. They do not buy skips; they buy Passes that boost their Coin and Shard income to ensure they can afford the massive 250,000 Coin costs waiting at the end of the timer.

- **End Game (5000+ Trophies): The Elder Skips.**

Tier 5 upgrades take 48 hours. At this stage, highly competitive players will drop \$19.99 for a 2,800 Gem Vault to bypass 48-hour timers (costing ~650 Gems per skip) right before a weekend tournament, driving the core ARPDau (Average Revenue Per Daily Active User) of the title.

Works cited

1. deep_economy_audit.md